

Bioimage analysis 27 years ago (1997)



- **ImageJ** (J for Java)---still the main game in town
 - Released in 1997 as rewrite of *NIH Image*
 - Apple in decline, the PC on the rise
 - 1995: Sun Microsystems created the Java programming language as an operating system–agnostic programming platform
 - Rasband wanted to support Windows, but not two sets of code
 - “*Instead of ‘write once, run everywhere’, Rasband found himself writing once and debugging everywhere.*”
- First autonomous coast-to-coast drive of the United States two years earlier

Bioimage analysis 12 years ago (2012)

- [AlexNet](#) is published (2012)
 - Wins the *ImageNet Large Scale Visual Recognition Challenge*
 - Error of 15.3%, more than 10.8 percentage points better than #2
 - Possible due to GPUs
- [CellProfiler](#) has been around for 6 years (2006)
- [Fiji](#) (Fiji is just ImageJ) has been around for 5 years (2007)
- [scikit-image](#) released 3 years ago (2009)
- Bitcoin grows from \$5 to \$13 (remember Silk Road?) (2012)
- U-Net, GANs, and [Jupyter](#) will appear in 2-3 years (2014/15)
- [AlphaGo](#) will beat Lee Sodol in 4 years (2016)
- [QuPath](#) is still 4 years in the future (2016)



CellProfiler™
cell image analysis software



scikit-image
image processing in python

